Saptarshi Mandal

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Personal website | Google Scholar

Education

University of Illinois Urbana–Champaign (UIUC) Expected graduation date: Spring, 2027
PhD in Electrical and Computer Engineering (ECE); GPA: 4.0/4.0
Advisor: Prof. R. Srikant

Indian Institute of Technology (IIT) Kharagpur Aug 2016 – Apr 2021
Joint B.Tech–M.Tech in Electronics and Electrical Communication Engineering (EECE) (major)
Minor in Computer Science and Engineering; GPA: 9.34/10

Research Interests

Sequential decision-making and reinforcement learning in large state/action spaces; distributionally robust RL and bandits; resource allocation and online learning; model compression including knowledge distillation; theory–practice bridge for RL/LLMs.

Publications & Preprints

- **S. Mandal**, Y. Murthy, R. Srikant. *Finite-Time Bounds for Distributionally Robust TD Learning with Linear Function Approximation*
  Under review at the International Conference on Learning Representations **(ICLR) 2026.**
  Preprint: arXiv:2510.01721
- **S. Mandal**, S.T. Kong, D. Katselis, R. Srikant. *Spectral Clustering and Labeling for Crowdsourcing with Inherently Distinct Task Types.*
  Transactions on Machine Learning Research **(TMLR), 2025.**
  Paper: Link
- **S. Mandal**, X. Lin, R. Srikant. *A Theoretical Analysis of Soft-Label vs Hard-Label Training in Neural Networks.*
  Learning for Dynamics and Control **(L4DC), 2025.**
  Paper:Published version; arXiv:2412.09579
- G. Xiong, H. Wang, Y. Pan, **S. Mandal**, S. Shah, N. Boehmer, M. Tambe. *Finite-Horizon Single-Pull Restless Bandits: An Efficient Index Policy For Scarce Resource Allocation.*
  International Conference on Autonomous Agents and Multiagent Systems **(AAMAS), 2025.**
  Paper: Published version; arXiv:2501.06103
- S. Mandal, G. Das. *An Online Delay-reliable Efficient Algorithm for Multi-resource Allocation.*
  Manuscript in preparation.

Research Experience

Graduate Research Assistant, UIUC ECE Aug 2021 – Present
Advisor: Prof. R. Srikant

Distributionally Robust TD and Q Learning with Linear Function Approximation 2024 – 2025

- Developed the first robust TD learning and Q-learning algorithms with linear function approximation, where robustness is measured using total-variation and Wasserstein-1 uncertainty sets without restrictive assumptions on the model, with a provable convergence guarantee.
- Proved non-asymptotic convergence guarantees that closely match non-robust counterparts, closing a key gap between empirical success of robust RL and existing theory.

Soft-Label vs Hard-Label Training in Neural Networks 2023 – 2024

- explanations for phenomena such as model compression via knowledge distillation in neural networks.

- Validation results using deep neural networks, with VGG 8+3 as the teacher and VGG 2+3 as the student.
- Accepted at L4DC 2025.

Inferring Labels from Multi-type Crowdsourced Data 2021 – 2022

- Extended the Dawid–Skene model to multi-type tasks and proposed a spectral clustering method to partition tasks into two groups by type; showed that perfect recovery is possible when the number of workers scales logarithmically with the number of tasks.
- Accepted at TMLR 2025 and presented (talk) at the INFORMS Applied Probability Society Conference.

Industry & Research Internships

Google Research — AI for Social Good Oct 2023 – Dec 2023

Supervisor: Prof. Milind Tambe (Harvard University); Collaborator: Niclas Boehmer (HPI, Germany)

- Explored data-driven approaches to efficiently allocate limited intervention resources and encourage patients to seek healthcare in rural India.
- Contributed to modeling rural healthcare outreach as a general restless bandit framework tailored to a large-scale intervention-effect dataset from Google Research India.
- Work accepted as a full paper at AAMAS 2025 (see above).

NVIDIA, Bangalore, India — Project Intern (Digital Profile) Summer 2019

- Automated fault aggregation and error collation from log data of the Orin SoC architecture by integrating Google Drive APIs into existing scripts and parsing IP metadata.
- Built a comprehensive latency calculator over the SoC architecture using graph-based shortest path algorithms and recursive methods in Python, achieving exponential reductions in latency computation time.

Teaching & Academic Service

Teaching Assistant, UIUC

- Graduate course: *MDPs and Reinforcement Learning (2025), Introduction to Optimization(2024)*

Teaching Assistant, IIT Kharagpur

- Undergraduate Lab: *VLSI Lab(2021), Microcontroller Systems Lab(2020)*

Awards & Honors

- IISc Scholarship "KVPY" (Kishore Vaigyanik Protsahan Yojana), 2016.
- IISc Summer Research Fellowship, Indian Academy of Sciences, 2020.

Technical Skills & Selected Coursework

Programming & ML: Python (advanced), PyTorch, Matlab, C, C++, Verilog.

RL & Simulation: Robust RL, OpenAI Gym, MuJoCo.

Optimization & Tools: CPLEX, Simulink, OMNeT++, Xilinx ISE, LaTeX, Beamer, Git.

Selected Graduate Coursework (UIUC; grades A/A+): Statistical Learning Theory, Statistical Reinforcement Learning, Neural Networks and Applications, Random Processes, High-dimensional Probability, Information Theory, Queueing Theory, Adaptive Systems, Network Optimization, Game Theory, Approximation and Online Algorithms, Communication Signal Processing and Algorithms.

Other Academic/Community Involvement

- Former Captain, Bengali Dramatics, Nehru Hall, IIT Kharagpur.
- Master of Ceremonies, main event of Bengali Student Organization, UIUC.

Last updated: November 24, 2025